NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE

Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED | **MATHEMATICS** | **M.M. 62**

REVISION ASSINGMENT # 10 | (FUNCTION & ITF) | TIME : 60 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. A polynomial function satisfies $f(x)f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$, $x\neq 0$ and $f(3)=-26$, then value of $f(4)$ is-

(A) –15 (B) –63 (C) –47 (D) –255

2. Let $f(x)$ be polynomial of degree 5 with leading coefficient unity such that $f(1)=5$, $f(2)=4$, $f(3)=3$, $f(4)=2$, $f(5)=1$, then $f(6)$ is equal to -

(A) 0 (B) 24 (C) 120 (D) 720

3. A and B are two sets such that n(A) = 4 and n(B) = 5. If number of possible mappings defined from set A to B is x and out of these y mappings are one-one, then x + y will be -

(A) 720 (B) 740 (C) 745 (D) 900

4. Range of the function $f(x)=\frac{2|x|-1}{3|x|+2}$ is-

(A) $\left(-\frac{2}{3},\frac{2}{3}\right)$ (B) $\left[-\frac{1}{2},\frac{2}{3}\right)$ (C) $\left[-\frac{1}{2},\frac{1}{2}\right]$ (D) $R-\left\{\frac{2}{3}\right\}$

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5. Domain of the function $f(x)=\frac{1}{x}+\frac{1}{\sqrt{x^2-4x}}+\log x$ is-

(A) $[4,\infty)$ (B) $(4,\infty)$ (C) $[0,4]$ (D) $(-\infty,4)$

6. If $\cot^{-1}\frac{n}{\pi}>\frac{\pi}{6}$, $n\in N$, then the maximum value of n is-

(A) 6 (B) 4 (C) 5 (D) 30

7. The value of $\sin^{-1}\left(\frac{1}{2}\right)+\cos^{-1}\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)+\tan^{-1}\left(2-\sqrt{3}\right)$ is-

(A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{5\pi}{12}$

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

8. Which of the following function(s) is/are odd function(s) ?

(A) $f(x)=\frac{x^3}{2^{|x|}+\sec x}$

(B) $f(x)=x^4e^{\left(\sqrt{x^2+1}+x^4\right)}$

(C) $f(x)=\frac{\left(e^x-1\right)}{\left(e^x+1\right)}\left(g(x)+g(-x)\right)$

(D) $f(x)=h(g(k(x)))$ when h(x) , g(x) & k(x) are all odd function

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9. If $f(x)=\sqrt{|x|^2-5|x|+6}+\sqrt{8+2|x|-|x|^2}$, then x can be-

(A) [–4,–3] (B) [–3,–2] (C) [–2,2] (D) [3,4]

10. $2\cot^{-1}7+\cos^{-1}\frac{3}{5}$ is equal to-

(A) $\cot^{-1}\frac{44}{117}$ (B) $\text{cosec}^{-1}\frac{125}{117}$ (C) $\tan^{-1}\frac{4}{117}$ (D) $\cos^{-1}\frac{44}{125}$

11. The value of $\cos(2\cos^{-1}x+\sin^{-1}x)$ at $x=\frac{1}{5}$ is -

(A) negative (B) positive (C) rational (D) irrational

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 12 to 14

Consider three functions $f(x)=x^2-4x$; $g(x)=x+5$, $h(x)=x-12$.

On the basis of above information, answer the following questions :

12. Number of integral values, which are not in the range of $\frac{g(f(x))}{h(f(x))}$ is -

(A) one (B) two (C) three (D) four

13. If the largest interval of x for which $g(f(x))$ is invertible is taken as $[a,\infty)$ and least value of $h(f(x))$ is 'b' then the value of $\sin^{-1}\left(\frac{8}{b}\right)+\cos^{-1}\left(-\frac{a^3}{b}\right)+\tan^{-1}\left(-\frac{32}{ab}\right)$ is -

(A) $\frac{2\pi}{3}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{12}$ (D) π

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14. Value of $\lim\limits_{x\to7}\lim\limits_{n\to\infty}\frac{[g(h(x))]^{2n}+1}{h(f(x))}$ is equal to -

(A) $\frac{4}{9}$ (B) $\frac{2}{9}$ (C) $\frac{1}{9}$ (D) Limit does not exist

SECTION–III

Numerical Grid Type (Single digit Ranging from 0 to 9) (4 Marks each, –1 for wrong answer)

1. Number of integer(s) in the range of $f(x)=\text{sgn}(\ell n(x^2-2x+2))$ is (where sgn(x) represents signum function)

2. Let $f(x)=\frac{1}{(1+\cot x)}$, then the value of $f(x)+f\left(\frac{\pi}{2}-x\right)+f\left(\frac{\pi}{2}+x\right)+f(\pi-x)$ is

3. Number of solutions of equation $\sin^{-1}(\sin6x)=-\frac{x}{2}$ in $[0,\pi]$ is

4. Number of integral value(s) of x in the domain of $f(x)=\sin^{-1}(x-1)+\log_\pi(2-x)$ is

REVISION ASSINGMENT # 09 DETERMINANT AND MATRIX MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	A	C	A	C	A	D	B,C,D	B,C
	Q.	11	12	13	14	15					
	A.	C	C	A	D	A					
SECTION-II	Q.	1	2	3	4						
	A.	7.00	6.00	25.00	162.00						

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